Structure and Enum C#.Net



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Overview



Introduction to Structure

How declare a structure?

Introduction to Enum

How to declare Enum?

Introduction to Structure

Structures

•Structures are value types and not reference types.

•They are similar to classes in many ways But as they do not require heap allocation and are allocated on a stack.

•A struct types can Declare constructors, constant, fields, methods, properties, operators and nested types.

•Classes and structures has many resemblances and three differences as given Below.

- Structures are Value Type and Classes are Reference Type.
- Classes can be inherited but Structure can not be Inherited.
- Classes fields can be initialized but Structure fields can not be initialized.
- Structure does not support parameter-less constructor.

How to declare a Structure?

Example:

```
struct Books
   {
      int ISBN;
      string AuthorName;
      string BookName;

public void Accept(int ISBN, string AuthorName, string
BookName)
      {
         this.ISBN = ISBN;
         this.AuthorName = AuthorName;
         this.BookName = BookName;
      }

      public string Display()
      {
          return string.Format("Book ISBN={0}
Author={1} and
          Name={2}",ISBN,AuthorName,BookName);
      }
   }
```

```
static void Main(string[] args)
      {
         Books mybook = new Books();
         mybook.Accept(123, "Donis Marsh", "C# 5.0");
         Console.WriteLine(mybook.Display());

         Console.ReadLine();
      }
```

Introduction to Enum

Enumeration

•An enumeration (enum) is a special form of value type. Which inherits from System.Enum and supplies as alternate name for an underlying primitive type.

•An EnumType has a name, underlying type and set of types(Int16,Int32,Int64).

•The fields are Static literal fields, each of which represents a constant.

How to declare an Enum?

Example:

```
enum Month
{
    January = 1,
    February,
    March,
    April,
    May,
    June,
    July,
    August,
    September,
    October,
    November,
    December
}
```

```
class Program
{
    static void Main()
    {
        int birthMonth = 7;
        Month month = (Month)birthMonth;

        Console.WriteLine($"Birth Month: {month}");
        Console.WriteLine($"Numeric value of Birth Month:
{birthMonth}");
    }
}
```

References

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